- If you forgot your 'from' while coding click it (CASE SOINTS)
  -> table
  -> ⊞ / : on table
  - copy and paste it

23rd June 26.

Topic: Case Statement SQL

-> It evaluates conditions and returns a value based on the first true condition
-> Way of building logical steps based on conditions
-> Allows to create a new column based on logical conditions applied on other columns
-> It allows us to classify our numerical data into categorical groups.

* It evaluets conditions and returns a value based on the first true condition

Syntax:

```
SELECT Column1,
    CASE
        WHEN Condition1 THEN result1
        WHEN Confition2 THEN result2
    ELSE    default_result
    END As  Column_name
    From    table_name
```

Employees

emp-id	name	salary
101	John	3000
102	Peter	7000
103	Samson	12000
104	Davus	4000
105	Goliyadi	300

numeric
Want 2 classify salary / High / Low

Sa

Business Logic

Salary below 5000 - Low range
Salary between 5000 and 10000 - medium range
Salary above 10k - High range

logical conditions

```
SELECT name, salary
CASE
    WHEN salary < 5000 THEN 'Low'
    WHEN salary Between AND 10k THEN 'Medium'
    WHEN Salary > >10k THEN 'High'
    END As salary_braket
    FROM employee
```

NB: The case statement is placed within the Select Statement
- Case Statements creat a new column

OUTPUT / Returns

name	Salary	Salary_bracket
John	3000	Low
Peter	7000	Medium
Samson	12000	High
David	4000	Low
Goliyadi	300	low

(On Exercise 3")

Products

Pro_Id	Pro_name	Price
1	Laptop	1200
2	Phone	800
3	keyboard	45
4	Monitor	300
5	Mouse	25

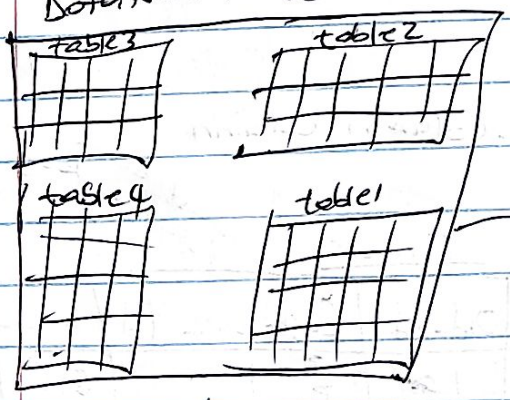
- Expensive if price >1000
- Mid rang if price between 100 and 1000
- Budget if price < 100

Product_name	Price	Price category
Laptop	12k	- Exp
Phone	800	- Mid range
Keyboard	45	budget
Mountor	300	mid-range
Mause	25	budget

```
SELECT Product name,
       Price
CASE
    WHEN Price > 1000 THEN 'EXPENSIVE'
    WHEN Price BETWEEN 100 AND 1000
    WHEN Price <100 THEN 'Budget'
    END As price_category
    FROM Products;
```

24th June SQL JOINT!!

-> Data Warehouse



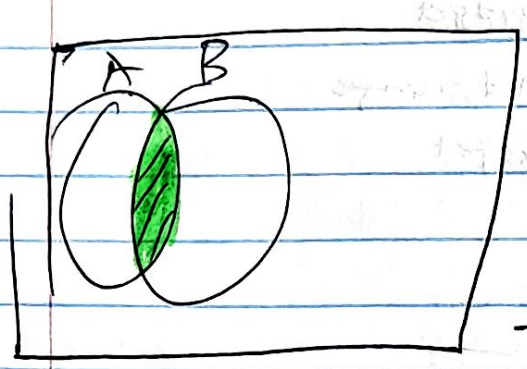
Database - plc where we store data in table
- central database
- Containg different content by they are related.
CRelational database
So we code in DMS - (databricks)

SQL Coding - then we apply the Joins with multiple tables

SQL Joins

- Meant to join multiple tables based on common columns.
- Must exist in both columns
-

INNER JOIN



- We join tables on column but the join affects the output rows

- When we join table we can select any column between the table we are joining

- Inner join - Returns rows where there is a match on both the tables

Syntax : SELECT A.column1
A.Column2
B.Column3
B.Column4;
FROM table1 AS A
INNER JOIN TABLE2 AS B
ON A.Comon_column = B.Column Column

Primary key

Customers

Cust_id	Cust_name
1	Rofhi
2	Mbalu

Orders (must be a foreign key)

Pro_Id	Ord_id	Customer_Id	Amount	Pid
131	101	1	500	Appl
132	102	2 (4)	20	Bag
133	103	3 (5)	10	Bag

↓ fact table

Write a SQL join to join 2 tables

```
SELECT *. Cust_Id,
       name
       amount
FROM Customers As A
INNER Join Orders As B
  ON A.Customer id = B. Customer id;
```

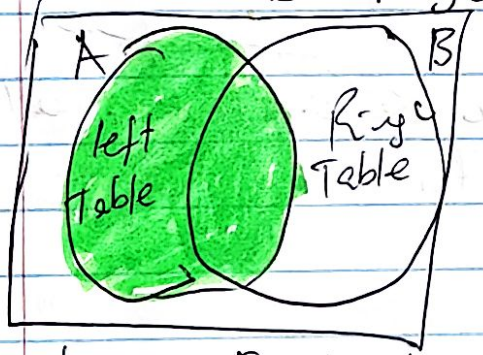
Customer Id	name	Amount
1	Rafhi	300

so; In this manner; the primary key a unique Identify e.g Customer Id in the custome table, becz its diffent for every Customers.

Prod_id	Product name	-Static descriptive Informatuon.
131	Apple	
132	mango	
13	Orang	

25th June

LEFT Join



In the Final Outcome

- Ja Join everything on the Left (all the rows) on the left table

* This Priority Table | Base Table | Main table Focus table
* Contains Critical Informatuon | data

Syntax : SELECT Column1
Column2
Column3
FROM table1 AS A
LEFT JOIN table2 AS B
~~ON A.Column~~
ON A common Column = B Comman_Colum

A Left Join returns all records from the left table, and Only the matching records from the right table. When there is no match in the right table we still keep the left table row, but the right table colum will show Null

Customers table

Cust_id	customer_name
1	Rfhwifa
2	Lerato
3	Mbali

Orders_table

Order_id	Cust_id	amount
101	1	500
102	2	300

Write a SQL Query to give me a list of all customers and what they order.

SELECT ~~Colu~~ cust_name,
Cust_id
amount
order_id
FROM Customer table1 AS A
LEFT JOIN order_table AS B
ON A.cust_id = B.cust_id

CustSid	cust name	Amount	order_id
1	Rafiwa	500	101
2	Levato	300	102
3	Mbali	Null	Null

only exist in one TABLE (A)

NULL - Missing Space
- Missing value

18 July

Null - empty space
- blank

SALES

Products	Amount
Apple	1000
Orang	NULL
Banane	200

-Find total Revenue

```
SELECT Sum(Amount) As Revenue
FROM Sales
```

1000 + 200 + NULL

IF Null (Statement) : to help us if we have Null
(Colum name) ——> Expression, replecement value

```
SELECT product, IF Null (Amount, 0) As Amount
FROM Sales
```

Returns:

Produc,	Amount
Apple	1000
Orange	0
BANANA	200

COALESCE( )

(Col1, Col2, Col3, Col4 --- Col12)
If 1st Column = Null Replace it with 2nd Column 2nd)

- Is a way to replace null values by a value from alternative Column or a specified value.
- We commonly use it in our SELECT Statement
- Order matter - from left to Right

e.g Customers table

cust-id	Cust name	phone 1	phone 2 (whats number
101	John	071 111 1111	NULL
102	Mary	NULL	082 222 2222
103	Peter	NULL	NULL
104	Abednigo	071 234 5678	082 543 2701

COALESCE

-Write an SQL Query to give me the name of the Custom and their contact details. (Replace null numbers with 0000)

```
SELECT  Cust_name,
        COALESCE(phone_number, whatApp_number, 0000) A.
                                  Contact number
FROM Customers;
```

• If you have phone, give, whats-A but of 0000

~~Cust-id~~	Cust_name	Custer_number	~~(whats-App number)~~
	John	071 111 1111	
	Mary	082 222 2722	
	Peter	0000	
	Abednigo	071 234 3678	

IS NULL: A way to check if there is Null value in a specific Column
- We can use it in our (WHER -statement)

e.g Customer table

Cust_id	Cust_name	numbe	What App:
101	John	071 111 1111	Null
102	Mary	null	082 222 2222
103	Peter	Null	null
104	Abe	071 234 5678	082 543 2101

```
SELECT name,
       phon-number
From table
WHERE phone- is Null ';
```

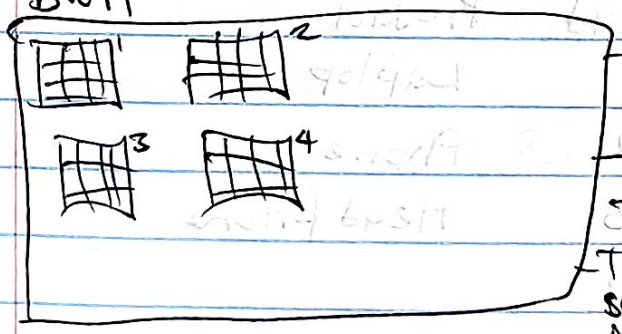
Returns

Customer	phone_number
Mary	null
Peter	null

14 July .26

Joins

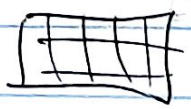
- We want to combine multiple tables into 1 single table
- We join tables based on a common column between the different tables
- DWH-



all related to one another by a specific column

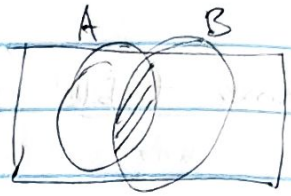
→ to dms (databricks)
→ Write SQL ~~Query~~ code

This will give joint summized joint table
Aggregates also



Inner Join

- 2 or more tables into 1 tables
- Returns ONly rows that have a matching values on both tables



Syntax:
```
SELECT col, col8, col3
from tableIAs A
Inner Joint table2 As B
ON A.Colum-Colum = B. common Column;
```

e.g

Customers

cust_id	Cust_name
1	Thabo
2	Sarah
3	Lerato
4	John

Orders

order_id	Cust_ID	Product
101	1	Laptop
102	2	phone
103	2	Headphone
104	5	Tablet

Who? What they Order?

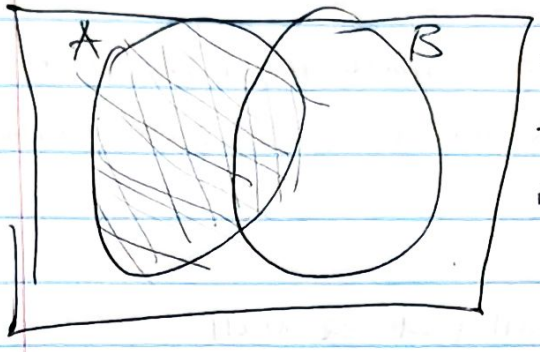
```
SELECT Custerm ID,
       Custumer_name,
       Order-Id,
       Product
FROM customers As A
Inner Joint Orders As B
ON A common-column ID = B. Customer ID's;
```

* to join MORE Use "AND"
e.g A cust_ID = B.Cust-ID AND
A.date_ID = B date_ID;

Returns=

Cust_id	Cust_naime	Order_id	Product
1	Thabo	101	Laptop
2	Sarah	102	Phone
2	Sarah	103	Head phone

LEFT JOIN



-Returns all the rows from the left table and only the matching rows from the right table

takes everything in the left table and all matching in the right table.

Syntax: SELECT Col1,
Col2,
Col3,
FROM table1 As A
LEFT JOIN table2 As B
ON A.Common_column = B.Common_column;

Customers

Cust_id	Cust_name
1	Thabo
2	Sarah
3	Lerato
4	John

Orders

Order_id	Cust_id	Product
101	1	Laptop
102	2	phone
103	2	Headset
104	5	Tablet

SELECT Cust_id,
Cust_name,
Products
From (table) Customer As A
Left Join Orders As B
On A.customer_id = B.Custome_id

Returns

Cust_id	Cust_n	Prod
1	Thabo	Laptop
2	Sarah	Phone
2	Sarah	Headph
3	Lerato	Null
4	John	NULL

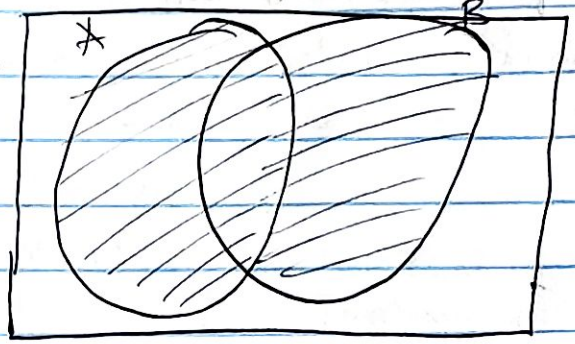
L v
Right User..

15 July

Full Outer Join

- Returns; all rows (matching) to combine them in one row.
All the row that are matching on common colums are combined into a single row.
- All unmatched row = Null
-> Where ther is no match we will put a Null
- Looks for everything in both table.

Example



```
SELECT  Col1, Col2, Col3
FROM table1 As A
FULL OUTER Join table2 As B
ON A common_column= B common_column;
```

E.g

Customer

Cuid	Cust_name
1	Thabo
2	Sarah
3	Lerato
4	John

Orders

Order_id	Customer_id	Product
101	1	Laptop
102	2	Phone
103	2	Headphones
104	5	Tablet

Syntax;

```
SELECT  Cust_id
        CustomerName
        Order Id
        Product
From Customers As C
Full Outer Join Orders As O
On C.Cid = O.Customer_ID;
```

Returns (C)

Cust_id	Name	Order_Id
1	Thabo	101
2	Sarah	102
2	Sarah	103
3	Lerato	Null
4	John	Null
5	Null	104

(removes duplicates) (doesn't Remove duplicates)

UNION and UNION ~~OR~~ ALL

- Combins Results of two SELECT statements by Stacking the rows → putting things atop of each other
- Order Matters on Syntax
- The SELECT / EACH SELECT statement must have the same # of Columns
- The Colums must be in the same order
- The columns must have the same data types

Stud = stu_id and student names

Current_ Student

Student_id	Student_name
1	Thabo
2	Sarah
3	Lerato

New_Student

Student_id	Student
3	Lerato
4	John
5	Mary

UNION

```
SELECT student_id, Student name
FRom Current_Students
UNIOWN
SELECT  Student_id,
        Student_name
FRoWEN New Student;
```

Returns

Students_id	Student name
1	Thabo
2	Sara
3	Lerato
4	John
5	Mary

* IF using UNION ALL = LERATO will Appear twice

```
SELECT Student_id, name
FROM Students
UNION All
SELECT student_id, name
FROM newStudents
```

20/Jul

Date fun

-> Command - Databricks

SELECT CURRENT_DATE() As today_dt;

-> gives today date

- SELECT CURREN_TIMESTAMP() As dt_time;

-> gives you current time

-- DATEDIFF() end_date, Start_date)

SELECT('2026-07-17', '2010-06-01') As dt_difference;

String

SELECT TO-DATE()

(String)

ment to convert smthn that is NOT a date, a date. Ensure that a date is read as a date)

TO-CHAR() As month_id;

-> To convert a date into a String

SELECT TO_CHAR ('2026-07-17', YYYYMMDD) As mon

To Convert (TO_DATE

date

-> (SELECT TO-DATE('2026-07-17'), TO_DATE(.

DATEDIFF

SELECT DATEDIFF('2026-07-05', '2026-07-01') As dt_difference;

end_date - start_date = Value

-- DATE_ADD (Measure, number of days / value)

-- Measure = DAY, MONTH, YEAR

SELECT DATE_ADD(DAY, 2026-07-05, '5) As add_5_days

SELECT DATE('2026-07-05' : -5) As Add_5_days

minis 5 days

Holy Xmas-Nwit-

SELECT DATE_SUB('2026-07-05', 2) As Rmoving_2_day;

SELECT TO_CHAR(TO_DATE('2026-07-17', 'MM') As_Month
SELECT TO_CHAR(TO_DATE('2026-07-17', 'DD') As day_value;

SELECT YEAR('2026-07-05') As year_dt;
Extracts - year from date

SELECT DAY('2026-07-05') As day_dt give a date
SELECT MONTH('2026-07-05') As month_dt; (gives you >7).

SELECT DAYNAME('2026-07-05') as day_name; Sunday

SELECT MONTHNAME('2026-07-05') As month_name;

SELECT DAYOFWEEK('2026-07-05') As day_name; (Sunday)?? 1st day of the week

SELECT ADD_MONTHS('2026-07-05', 2) As adding_2_month;
adds 2 months to the date +2 = 2026-09-05

EDA

21st June - CTE -

-- Creating a database in databricks..!
TABLE OPARATION - Query to create table raw data and Perform multiple operations - How they work.
-> Create a Catalog
Called Fnx: CREAT CATALOG IF NOT EXISTS June_intake; ✓
Go to Catalog My Organization workspace
* It creat database within the warehouse
"June_Intake.

- USE CATALOG Jun_Intake; Run code

Just Create Catalog mbalidbabe;
Drop Catalog IF exist mbali_babe; CASCADE

Wand to schema

6. CREAT SCHEMA IF NOT EXISTS Student_profile;
-> it creats a schema unfo warehouse

Whats inside schema: tables -> DROP SCHEMA Student_Profile; is to delay

CREAT OR REPLACE TABLE June-Intake.Student-P

-> this code is used to create a schema with data base

fx: CREAT OR REPLACE TABLE June-INtake.Student, Profile (

Student_id INT,
name STRING,
Surname STRING,
age INT
email STRING
registration_DT DATE);

SELECT *
FROM June_Intakes Students, Profile;
will only Return empty columns;

fx: INSERT INTO June-intake.Studen, profile Value
Value
(101, 'Rochester', Jowe, 28, 'rochester@google.com', 2019-0
(102, 'Sive', 'Magazi', 33, 'sive@phalaphala.com' '1912-01-01',
(103, 'Cassper', 'Zuma' 73, 'Czuma@NKanta.com' '2020-01-12',
(104, 'Mbali, 'Nhlapho, 41, 'mbali@cleaning.com' '2010-10-12')

SELECT *
From June-intake, Student, Profiles;

fx: DELET
FROM June-intake. Students. Profiles
WHERE age<50;
(deleted Zuma)

delete = rows, contents based on conditions
drop down = involves erasing the structural container or database itself

We use delete function to remove a specific row from data based on specific condition

le fx; UPDATE;

```
UPDATE June-INTake. Students. Profile
SET age = 25
WHERE Student-id IN (101);

SELECT *
FROM June_INTake.Student. profile;
- will change
(deleting a column):
ALTER TABLE June_intake. student. Profiles
DROP. Column registration_DT;
```

22 July. 26

```
NEW - script. - use same table
SELECT * FROM June_Intake-Student profile;
- Create or Replace TEMPORARY TABLE elders AS (
SELET* name surname
  FROM _June. take S. Profile);
SELECT* WHERE age>50);
FROM Elders;
```

temp- Saves session while coding
- To simply Complex queays
- To improve performance
- To clean and transform data

Ctel = I combine script long + 2 more tables,
temp = smaller part

SELECT*
FROM June
(elders)

CE1 - With ctel As(
SELECT name,
Surname
age
From June_Intake
WHERE age>40), < sales As (select * FROM sales).
SELECT*
FROM elders;

* With Olders As (.
SELECT name
Surname
age
From June
where age>4),
Sales As (
Select date As purchase_st,
gender, total Amount
From sales
Where Gender like('male')

From elders
-Temp_table-
Exist in session of coding - Re-Run Again

23rd July

Does not include catalog and schema name
cos it dissapears everytime you log out, Only exist
while coding.